J. Acoust. Soc. Jpn. (*E*) **11**, 1 (1990)

Acoustical properties of porous materials —Modifications of Delany-Bazley models—

Yasushi Miki

Faculty of Engineering, Takushoku University, 815–1, Tatemachi, Hachioji, 193 Japan

(*Received 11 March 1989*)

The characteristic impedance and propagation constant of porous materials are discussed from a viewpoint of physical realizability, and new regression models based on experimental data by Delany and Bazley are derived. Modifications from their original models are so made that the impedance function satisfies the positive-real property and that the propagation constant written in terms of complex variables becomes a regular function in the right half-plane. The new models are shown to be useful for the prediction of acoustical behavior of porous materials, especially in the double-layer case, even outside the frequency range of validity of the original models.

PACS number: 43. 55. Ev

1. INTRODUCTION

Acoustical properties of porous materials are specified by the characteristic acoustic impedance and the propagation constant. The experimental model for each value was given by Delany and Bazley,[1] and it has been used widely to predict an acoustical behavior of a ground surface. In the case of a multiple layer, however, the real part of the surface impedance when computed from their models sometimes becomes negative at low frequency range. This phenomenon is due to the fact that the existing impedance must be a positive-real function whereas that of their model cannot satisfy this property. In this paper, new models for the impedance and propagation constant are presented. They are the modifications of Delany-Bazley models and satisfy the positive-real property.

2. ELECTRICAL ANALOGY OF POROUS MATERIALS

Porous materials are usually considered as an assembly of capillary tubes. A single cylindrical tube is described equivalently by an electric circuit shown in Fig. 1.[2] If time-dependence $\exp(j\omega t)$ is assumed, each electric element is given as

$$L=\frac{\rho_0}{A}, \tag{1}$$

$$C=\frac{A}{\rho_0 c_0^2}, \tag{2}$$

$$R=\frac{S}{A^2}\sqrt{\frac{\omega\rho_0\mu}{2}}, \tag{3}$$

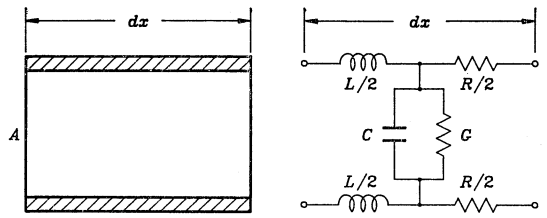
$$G=S\frac{\eta-1}{\rho_0 c_0^2}\sqrt{\frac{\lambda\omega}{2C_p\rho_0}}, \tag{4}$$

where A is the area of the tube, S the circumference of the tube, ρ_0 the density of air, c_0 the speed of sound, μ the viscosity coefficient, λ the coefficient of heat conduction, C_p the specific heat of air at constant pressure, and η the ratio of specific heats. Note that the resistance R and the conductance G are both proportional to $\sqrt{\omega}$ at high frequencies, and that R converges to a constant value

$$R_0=\frac{8\pi\mu}{A^2} \tag{5}$$

as $\omega\to 0$ (Poiseuille's law). This value is called the flow resistivity of a tube.

The electric circuit shown above is known as a two-port network. It is of course passive and linear.



(a) Cylindrical tube (b) Equivalent circuit

Fig. 1 Cylindrical tube of area A and its equivalent circuit.

Defining

$$y=G+j\omega C, \tag{6}$$

$$z=R+j\omega L, \tag{7}$$

we can write the characteristic impedance Z and the propagation constant γ of the tube as

$$Z=\sqrt{\frac{z}{y}}, \tag{8}$$

$$\gamma=\sqrt{zy}, \tag{9}$$

respectively. If we assume Eq. (5) and $G=0$, Eqs. (8) and (9) can be expressed as

$$Z=\frac{\rho_0 c_0}{A}\sqrt{1+\frac{AR_0}{j\omega\rho_0}}, \tag{10}$$

$$\gamma=\frac{j\omega}{c_0}\sqrt{1+\frac{AR_0}{j\omega\rho_0}}, \tag{11}$$

respectively.

Porous material in general may be described as a combination of similar but more complicated two-port networks. If a material is homogeneous and has infinite thickness, it is described as a one-port network composed of an infinite number of two-port networks in cascade connection. The impedance of this one-port network represents the surface impedance of the material.

One of other factors which affect acoustical properties of porous materials may be a porosity. However, we restrict our discussions to material whose porosity is near unity since Delany and Bazley gave experimental data only for those materials.

3. NEW MODELS

3.1 Positive-Real Property of the Impedance Function

The impedance function of a linear passive one-port network must be a positive real function.[3] The model derived by Delany and Bazley,[1] however, cannot satisfy that property. If the impedance function is of the form

$$Z(f)=R(f)+jX(f), \tag{12}$$

the positive-real conditions are given as follows:

$$Z(-f)=Z^*(f), \tag{13}$$

$$R(f)\geq 0, \tag{14}$$

$$X(f)=H[R(f)], \tag{15}$$

where * denotes the complex conjugate and $H[\cdot]$ means Hilbert transform. For Delany-Bazley type functions

$$R(f)=1+af^b, \quad X(f)=cf^d \quad (f>0), \tag{16}$$

the condition (15) can be satisfied if

$$d=b, \quad c=-a\cot\frac{(b+1)\pi}{2}, \quad -1<b<0. \tag{17}$$

Here, we used the relation

$$H[|x|^{\nu-1}]=-\cot\frac{\nu\pi}{2}\,\mathrm{sgn}(x)|x|^{\nu-1}, \tag{18}$$

and the condition (13) was assumed. These results also satisfy the condition (14) if $a\geq 0$.

3.2 Property of the Propagation Constant

Let $\gamma(s)$ be the propagation constant of a porous material. The transfer function

$$G(s)=e^{-\gamma(s)} \tag{19}$$

is a regular function in the right half of the complex s-plane. Therefore, the function $\gamma(s)$ is also regular in the same region if $G(s)\neq 0$. Assuming a single power-law relation for $\gamma(s)$, we can write

$$\gamma(s)=\frac{s}{c_0}(1+as^b). \tag{20}$$

Substitution of $s=j\omega(\omega>0)$ into Eq. (20) yields

$$\begin{aligned}\gamma(j\omega)&=\frac{j\omega}{c_0}(1+aj^b\omega^b)\\&=\frac{\omega}{c_0}\left[\left(-a\sin\frac{b\pi}{2}\right)\omega^b+j\left(1+\left(a\cos\frac{b\pi}{2}\right)\omega^b\right)\right].\end{aligned} \tag{21}$$

According to Delany and Bazley, we can rewrite Eq. (21) as follows:

$$\gamma(f)=\alpha(f)+j\beta(f), \tag{22}$$

$$\alpha(f)=\frac{\omega}{c_0}pf^r, \quad \beta(f)=\frac{\omega}{c_0}(1+qf^r) \quad (f>0). \tag{23}$$

The relation between p and q is expressed as

$$p=-q\tan\frac{r\pi}{2}. \tag{24}$$

Thus, the real and imaginary parts of the propagation constant cannot be determined independently.

3.3 Modifications of Delany-Bazley Models

The constants for the normalized impedance shown in Eq. (16) were given by Delany and Bazley as

$$a=9.08,\quad b=-0.75,\quad c=-11.9,\quad d=-0.73, \tag{25}$$

and those for the propagation constant shown in Eq. (23) as

$$\left.\begin{aligned} &p=10.3,\quad q=10.8,\\ &r=-0.59\ (\text{for }\alpha(f)),\quad -0.70\ (\text{for }\beta(f)). \end{aligned}\right\} \tag{26}$$

It is clear that Eqs. (25) and (26) do not satisfy Eqs. (17) and (24), respectively. Using experimental data by Delany and Bazley, we make new regression models subject to the constraint conditions (17) and (24). New sets of values are given as follows:

$$a=5.50,\quad c=-8.43,\quad b=d=-0.632, \tag{27}$$

$$p=7.81,\quad q=11.41,\quad r=-0.618. \tag{28}$$

The results are shown in Fig. 2 in comparison with the original ones.

3.4 The New Models in SI Units

We formulate the new models in SI units.

(1) The characteristic impedance normalized with respect to air:

$$Z(f)=R(f)+jX(f), \tag{29}$$

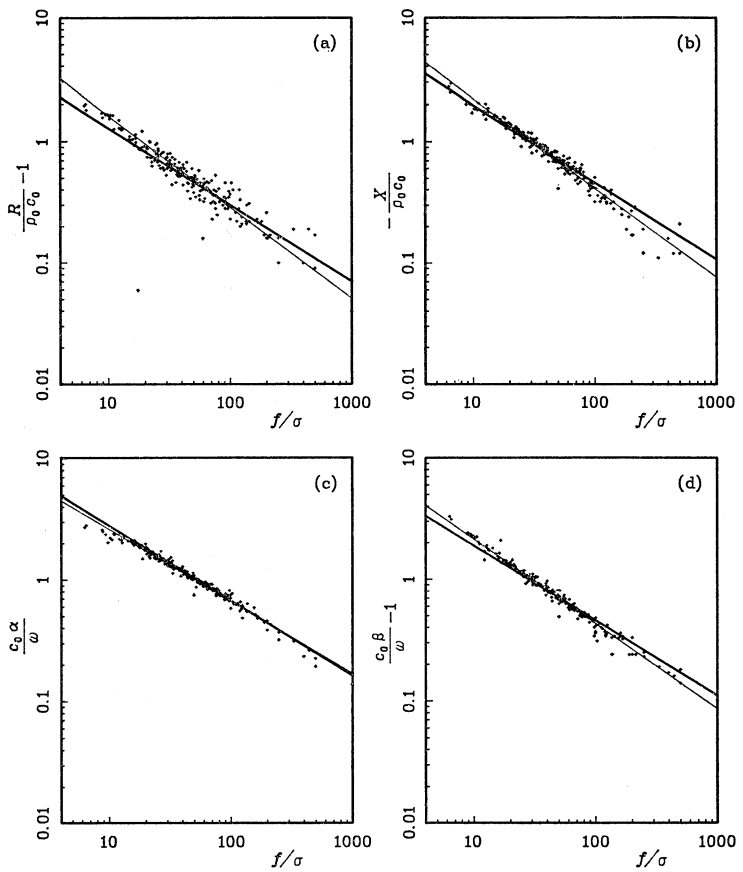


Fig. 2 New regression models for the characteristic impedance and propagation constant (thick lines). Thin lines show the original models. Dots show the experimental data by Delany and Bazley.

$$R(f)=1+0.070\left(\frac{f}{\sigma}\right)^{-0.632}, \tag{30}$$

$$X(f)=-0.107\left(\frac{f}{\sigma}\right)^{-0.632}. \tag{31}$$

(2) The propagation constant:

$$\gamma(f)=\alpha(f)+j\beta(f), \tag{32}$$

$$\alpha(f)=\frac{\omega}{c_0}\left[0.160\left(\frac{f}{\sigma}\right)^{-0.618}\right], \tag{33}$$

$$\beta(f)=\frac{\omega}{c_0}\left[1+0.109\left(\frac{f}{\sigma}\right)^{-0.618}\right]. \tag{34}$$

In above equations, σ is the flow resistivity of a material.

4. IMPEDANCE OF DOUBLE LAYER

4.1 Comparison between Original and New Models

Consider a rigidly backed layer. The surface impedance is given as

$$Z(f)=Z_0(f)\coth\gamma(f)l, \tag{35}$$

where $Z_0(f)$, $\gamma(f)$ and l are the characteristic impedance, the propagation constant and the thickness of the layer, respectively. Figure 3 shows the normalized surface impedance computed from the original models (thin lines) and the new models (thick lines) with $\sigma=100{,}000$ MKS units and $l=10$ mm. It is seen that these results show a distinct difference at frequencies below 1 kHz and that the real part of the former becomes negative with decreasing frequency. As stated above, such an impedance cannot be realized. This frequency range is found to be coincident with the range $f/\sigma<0.01$ (σ in MKS units) to which Delany and Bazley gave a warning not to extrapolate their models. Although it has not been verified yet that the new models provide a good prediction of the acoustic behavior in this frequency range, it can be said at least that the new models are well behaved in that range. Some examples of model fitting will be shown in the next section.

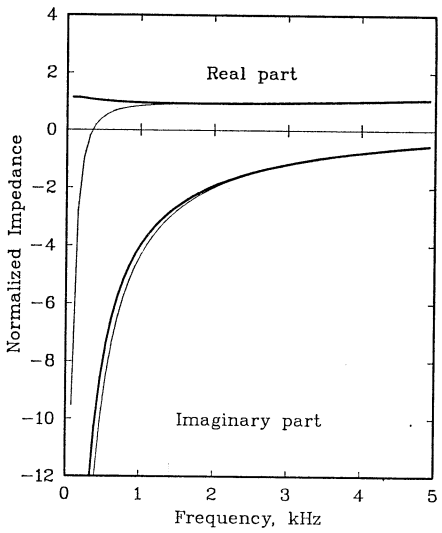


Fig. 3 Comparison between the surface impedance predicted from the new model and that from the oridinal model with $\sigma=100{,}000$ MKS units and $l=10$ mm.

4.2 Model Fitting for Experimental Results

Figure 4 shows the normalized surface impedance data obtained by C. Suzuki, H. Yano and H. Tachibana[4] for a rigidly backed glass wool of 50 mm thickness with 32 kg/m³ density. The thick curves show the impedance computed from the new models with $\sigma=9{,}000$ MKS units. The value of σ was so determined that the curves provide the best fit in the range $f/\sigma>0.02$ or $f>200$ Hz. The thin curves show the impedance computed from the original

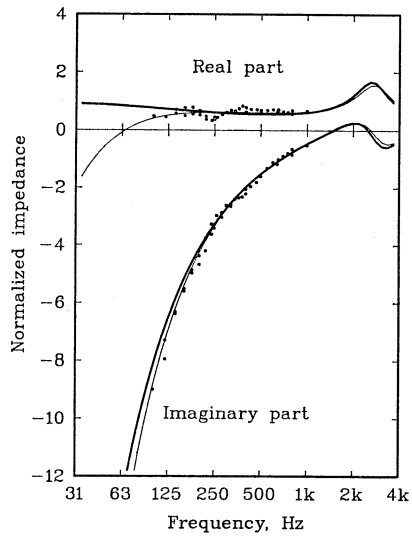


Fig. 4 Comparison between measured and predicted surface impedance. Dots are the measured data by Suzuki *et al.* Thick curves and thin curves show the predictions from the new model and the original, respectively, with $\sigma=9{,}000$ MKS units and $l=50$ mm.

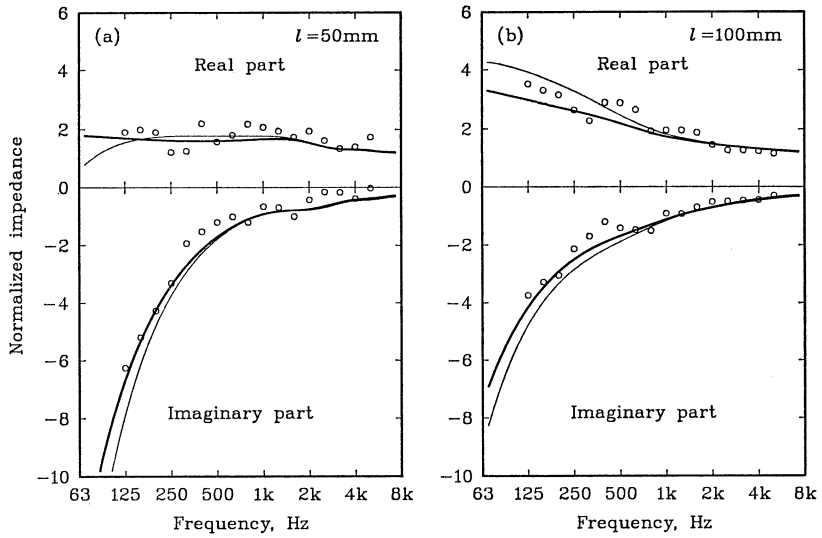


Fig. 5 Comparison between measured and predicted surface impedance. Dots are the measured data by Nakajima *et al.* Thick curves and thin curves show the predictions from the new model and the original, respectively, with σ=41,000 MKS units and (a) l=50 mm, (b) l=100 mm.

models with the same σ-value. It seems that the latter curves show the better agreement. Since all of the data lie in the range of validity of the original models, the acoustical behavior outside that range remains unverified.

Figure 5 shows the normalized surface impedance data obtained by T. Nakajima and T. Kawai[5] for a rigidly backed rockwool of 50 and 100 mm thickness with 150 kg/m^3 density. The thick curves in each figure show the impedance computed from the new models with σ=41,000 MKS units. The σ-value was so determined that the curves provide the best fit over the whole frequency range of measurement, and equal values was obtained from either set of data (Fig. 5(a) and (b)). The thin curves show the impedance computed from the original models with the same σ-value. If the σ-value is determined from data in the range $f/\sigma > 0.02$ or $f > 800$ Hz, it varies from 38,000 to 40,000 MKS units according to models and data sets. Since it differs little from the value σ=41,000 MKS units, this value was adopted as a typical value of the flow resistivity of the material. In this example, it may be said that the prediction from the new models is superior to that from the original over the whole frequency range, especially in the range $f < 400$ Hz, which is outside the range of validity of the original model.

5. CONCLUSIONS

The characteristic impedance and propagation constant of porous mateirals were considered from a viewpoint of physical realizability, and the new regression models based on the experimental data by Delany and Bazley were derived. It may not be expected that single power-law relations provide a perfect prediction of acoustical behavior over the entire frequency range. However, if the single power-law relation is used for the impedance function, it must be at least a physically realizable one. Otherwise it may cause significant errors in the low f/σ range. The new models thus obtained are found to be useful for the prediction of acoustical behavior even outside the range of validity of the original model. Further modifications for the porosity and the structure factor[6] will be discussed in a future paper.

ACKNOWLEDGMENTS

The author would like to express his gratitude to Dr. M. E. Delany for supplying experimental data measured 20 years ago. Without these data the new models would not have been proposed with

confidence. Thanks are also due to Prof. H. Tachibana and Mr. T. Nakajima for supplying experimental results.

REFERENCES

1) M. E. Delany and E. N. Bazley, "Acoustical properties of fibrous absorbent materials," Appl. Acoust. **3**, 105–116 (1970).
2) J. L. Flanagan, *Speech Analysis Synthesis and Perception* (Springer-Verlag, Berlin, 1965), p. 31.
3) L. Weinberg, *Network Analysis and Synthesis* (McGraw-Hill, New York, 1962), p. 244.
4) C. Suzuki, H. Yano, and H. Tachibana, "A new method of measuring normal incident sound absorption characteristics of materials using acoustic tube," J. Acoust. Soc. Jpn. (E), **2**, 161–167 (1981).
5) T. Nakajima and T. Kawai, "Measurement of acoustic impedance ratio of ground by standing wave method," Proc. Autumn Meet. Acoust. Soc. Jpn., 231–232 (1980) (in Japanese).
6) Y. Miki, "On the acoustical properties of porous materials," Tech. Rep. Inst. Electron. Commun. Eng. **EA81**-68, 7–14 (1982) (in Japanese).